IBCP YEAR 1 / ARTIFICIAL INTELLIGENCE / SUMMATIVE ASSESSMENT

Scam Shield AI

Human-centered fraud awareness, designed toward the bilingual goal in the assessment brief

STUDENT	PLATFORM	DELIVERABLE
Aarav Motwani	Google Dialogflow ES	Agent, tests and project documentation

01 The problem

People receiving urgent payment requests, fake job offers, impersonation calls, and unfamiliar links may act before checking who contacted them. Scam Shield AI gives clear, plain-language precautions and a route to official help. It does not claim to verify that a particular message or sender is genuine.

02 Audience and design goals

The intended users are digital-payment users, students and job seekers, older adults, and anyone who may already have shared information or transferred money. The response design prioritises a quick warning, a short explanation, and a safe next action. It avoids asking users to share real OTPs, UPI PINs, passwords, CVVs, or full account information.

Language requirement: The assessment brief calls for a bilingual assistant. The current live Dialogflow ES prototype is configured in English; a second language and bilingual test evidence are not yet implemented. This is an open requirement, not a completed feature.

03 Solution architecture

The agent package contains **21 intents** including welcome and fallback, **10 custom entities**, and **240 example training phrases**. Custom intents have 12 phrases each. A required requested-action parameter demonstrates slot filling; short-lived contexts support a scam-detail follow-up and an OTP-safety quiz. Quick replies give users obvious next steps. Responses run without a webhook or personal-data store.

USER NEED	INTENT GROUP	SAFETY OUTCOME
Check a suspicious message	ScamCheck, ScamDetails	Pause action; describe the request without sharing secrets
Understand a scam pattern	OTP, UPI, QR, phishing, job, investment, loan, impersonation, remote access	Recognise warning signs and verify independently
Recover and report	AccountCompromised, ReportFraud, PreserveEvidence	Contact bank quickly; use official reporting routes
Learn and navigate	SafePaymentTips, quiz, welcome, fallback, goodbye	Practice a safe response and recover from unclear input

DESIGN DETAIL

04 Language understanding

The custom entities extract the scam type, contact channel, requested action, credential type, payment method, claimed identity, urgency cue, suspicious-link feature, job fee, and whether harm already occurred. Synonyms cover common variations such as 'one-time password', 'UPI collect request', and 'registration fee'. The agent responds conservatively even if a phrase is misclassified.

ENTITY	EXAMPLE VALUES	WHY IT MATTERS
@scam-type	UPI fraud, phishing, job scam	Routes to relevant advice
@requested-action	share OTP, send money, open link	Supports required follow-up slot
@credential-type	OTP, UPI PIN, password	Highlights secrets never to disclose
@payment-method	UPI, QR code, card	Explains payment-specific risk
@contact-channel	WhatsApp, SMS, call	Captures where contact happened
@impersonated-identity	bank, employer, support	Identifies authority claims
@urgency-cue	immediately, account will close	Recognises pressure tactics
@link-feature	shortened link, misspelled domain	Recognises suspicious links
@job-fee	registration fee, deposit	Flags upfront-payment requests
@loss-status	money sent, app installed	Prioritises urgent recovery

05 Example conversation

STEP	USER / AGENT
1. User	Can you check if this message is a scam?
2. Agent	I cannot verify a message as safe from a description alone. Do not click links, pay, or share codes. What did they ask you to do?
3. User	They want me to share my OTP.
4. Agent	That is a warning sign. Do not share the OTP. Contact the organisation using its official app or website, not the sender's link or number.

06 Safety boundaries

The bot is advisory, not an automated fraud detector or a bank representative. It never asks for a real secret or claims a transaction is safe. If money has been lost, it directs the user to contact their bank immediately and use India's official financial cyber-fraud helpline **1930** and **cybercrime.gov.in**. Advice follows NPCI and RBI fraud-awareness guidance.

QUALITY AND HANDOFF

07 Test protocol

A live Dialogflow ES console pass covered ten cases: five scam-scenario intents, remote access, a slot/context follow-up, both quiz branches, and fallback. Exact observations are in the companion CSV.

SCENARIO	EXPECTED MATCH	EXPECTED BEHAVIOUR
UPI collect request	UPICollectIntent	Pass - matched UPICollectIntent; extracted payment method UPI; safe warning returned
OTP impersonation	OTPIntent	Pass - matched OTPIntent; extracted credential type OTP; advised never to share it
Fake job fee	FakeJobIntent	Pass - matched FakeJobIntent; registration-fee warning returned
Phishing link	PhishingLinkIntent	Pass - matched PhishingLinkIntent; extracted shortened link and SMS; advised not to open it
Remote access	RemoteAccessIntent	Pass - matched RemoteAccessIntent; warned against unsolicited remote-control app and advised recovery steps
Money already lost	AccountCompromisedIntent	Pass - matched AccountCompromisedIntent; extracted money-sent status; bank
Scam detail slot filling	ScamCheckIntent -> ScamDetailsIntent	Pass - ScamCheckIntent opened follow-up context; ScamDetailsIntent captured requested action share OTP and returned safe advice
Quiz, correct answer	QuizStartIntent -> QuizSafeAnswerIntent	Pass - No matched QuizSafeAnswerIntent; confirmed never share OTP and use official bank contact
Quiz, unsafe answer	QuizStartIntent -> QuizUnsafeAnswerIntent	Pass - Yes matched QuizUnsafeAnswerIntent; corrected answer and advised never share OTP
Fallback	Default Fallback Intent	Pass - matched Default Fallback Intent; safe options and 1930 guidance returned

08 Review and submission status

The Dialogflow agent is present as ScamShieldAI (agent ID scamshieldai-k9we), with 21 intents and 10 custom entities. Ten English console test cases passed. The bilingual requirement remains open. Web Demo could not be enabled during this pass because the Integrations page returned an IAM permission warning. A private GitHub repository has been created at github.com/aarav343/scam-shield-ai-sa; only the owner and invited collaborators can view it. Candidate registration number was not provided; fill it in only if required before student submission.

09 Sources

SA brief: CRS AI Y1C1 Artificial Intelligence - Real-World Applications and Implications (Summative Assessment).

Google Cloud Dialogflow ES documentation: agent import/export and intent parameters.

National Payments Corporation of India: fraud awareness. Reserve Bank of India: digital banking safeguards.

National Cybercrime Reporting Portal: cybercrime.gov.in and financial cyber-fraud helpline 1930.